

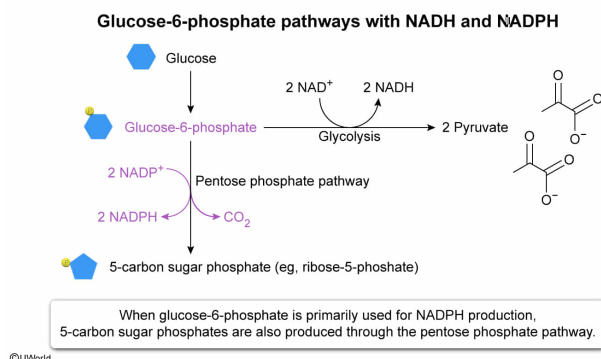
If glucose-6-phosphate is used for NADPH synthesis more than for NADH synthesis inside a cell, which of the following metabolic processes is most active in the cell?

- ☐ A. Reduction of glucose metabolites in the cytosol to produce ATP [7%]
- ☐ B. Reduction of antioxidant NADP^+ molecules [26%]
- ☒ C. Production of five-carbon sugar phosphates in the cytosol [60%]
- ☐ D. Oxidation of glucose metabolites inside mitochondria to produce CO_2 [5%]

Correct

60% answered correctly

Explanation:



The **pentose phosphate pathway** consists of two sets of reactions:

- The oxidative phase, a series of reactions in which two molecules of **NADP^+** are **reduced to NADPH** as **glucose-6-phosphate** is **oxidized to ribulose-5-phosphate**, a **five-carbon sugar phosphate** (ie, a pentose phosphate).
- The nonoxidative phase, during which ribulose-5-phosphate is isomerized to ribose-5-phosphate, which can be converted to various sugars.

The sugar phosphates and NADPH molecules produced in the pentose phosphate pathway are essential for normal cellular function. **NADPH participates in antioxidant reactions and donates electrons during reduction reactions** (eg, **fatty acid synthesis** and conversion of ribose to deoxyribose). **Ribose-5-phosphate** is the **precursor** for the **ribose sugars** found in **nucleotides** and can also be converted to intermediates in the **glycolytic pathway**.

The question asks for the most active metabolic process when use of glucose-6-phosphate for NADPH synthesis is greater than its use for NADH synthesis inside a cell. **Greater use of glucose-6-phosphate for NADPH synthesis than for NADH synthesis** indicates that **production of pentose sugar phosphates**

is the **most active metabolic process** among those listed.

(Choice A) During glycolysis, glucose metabolites are *oxidized* (not reduced) to produce ATP. However, if most glucose-6-phosphate in the cell is directed toward NADPH synthesis, then the pentose phosphate pathway is more active than the glycolytic pathway.

(Choice B) In the pentose phosphate pathway, NADP^+ is the precursor for NADPH. In the process, NADP^+ accepts electrons (ie, is reduced) and therefore acts as an *oxidant*.

(Choice D) Oxidation of glucose-derived carbons in the mitochondria to produce CO_2 would require glycolysis. Because glucose-6-phosphate is used more for NADPH production, fewer glucose-derived carbons are available to produce CO_2 in the mitochondria.

Educational objective:

The pentose phosphate pathway generates ribose-5-phosphate and NADPH. The NADPH is generated by the reduction of NADP^+ and serves as an electron donor. Ribose-5-phosphate is a precursor for various sugars, including the ribose used in nucleotide synthesis.

Time spent:0 sec

Subject:
Biochemistry

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System:
Metabolic Reactions